

An Independent Reproduction of the AYO ExoEarth Yield for the Habitable Worlds Observatory: Two Questions on Which It Turns and an application to a mass-selected planet sample in a narrowed habitable zone

Prepared in the `yields` vaibify project

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Abstract

Yield estimates for the Habitable Worlds Observatory (HWO) rest on the Altruistic Yield Optimization (AYO) framework of Stark et al., and the exoEarth candidate (EEC) counts that flow from it now carry architectural weight. We report an independent reimplementa- tion of that model, built from the published equations and parameter tables rather than from AYO itself, and use it to evaluate a counterfactual planet selection: planets of $0.5\text{--}2\,M_{\oplus}$ with $R = M^{0.27}$ in a habitable zone of $0.96\text{--}1.20$ AU scaled as $\sqrt{L_{\star}}$, against the canonical $R \leq 1.4\,R_{\oplus}$ planets in $0.95\text{--}1.67$ AU. The reimplementa- tion carries no fitted parameter: with the extended-source throughput treated as a function of separation, as Stark et al. (2019) define it, the uncalibrated 6 m yield is 20.92 against a published 22.5, and the throughput factor the pipeline formerly carried has been removed ($\kappa = 1.000$). Of 24 independent comparisons against the published record, 22 agree; the realized-yield distribution matches the published mean (21.0 against 21.0), median (18 against 18) and peak (11 against 10).

Two disagreements survive every attempt we have made to remove them, and we argue that both are under-determination in the published description rather than defects we can locate. First, spectral characterization times in our model depend on target difficulty and aperture far more steeply than Stark’s: the mean over the first eighteen candidates is 14.3 d at 6 m against a published 22 d. A 56-point factorial design over the four throughput-family parameters that could plausibly absorb this shows that *no* point in that family reaches the published value: among the 20 design points that reproduce the published yield to 10%, the characterization time spans only 7.0–11.6 d. Second, the $\eta_{\oplus}$ interval of Stark et al. (2024) Sec. 3.6 admits two readings, and a 2×2 decomposition attributes 96.1% of our residual disagreement with their Fig. 10 to that single choice. Following every documented choice simultaneously makes the agreement *worse*, not better (peak 15 against 10).

Each discrepancy reduces to one question that only the AYO authors can answer, and we state both explicitly. Until they are answered the reproduction is an “almost”, and we quantify what “almost” costs. The counterfactual result is meanwhile robust to both: the yield ratio is $0.239^{+0.030}_{-0.023}$, and across seven model variants spanning both readings it moves by less than 8%.

Contents

1	Introduction	2
2	An independent implementation of AYO	3
2.1	Scope and sources	3
2.2	The model as a graphical model	3

2.3	The nodes	3
2.4	What we do not calibrate	3
3	Agreement with the published record	5
4	First discrepancy: the characterization time budget	6
4.1	The symptom	6
4.2	The throughput family cannot supply the difference	6
4.3	What this implies	7
5	Second discrepancy: the reading of the occurrence interval	8
5.1	One sentence, two readings	8
5.2	A factorial decomposition	8
5.3	Following every documented choice makes the agreement worse	8
5.4	Testing the interval against its own source	9
6	The counterfactual, and its robustness	10
7	Discussion: the two questions	10
8	Conclusions	12
A	Verification against the published record	13
B	Published figures beside their model counterparts	13
C	Ledger of hypotheses tested	13
D	Every contributing star	13

1 Introduction

The Astro2020 decadal survey recommended a mission capable of surveying ~ 25 potentially habitable worlds. Translating that recommendation into an aperture, a coronagraph and a set of detector requirements runs through a yield model, and for HWO that model is AYO (Stark et al. 2014; Stark et al. 2019; Stark et al. 2024). AYO is not a simple scaling: it allocates exposure time across a target list by an equal-slope rule, so its output responds to changes in the planet population in ways a proportionality argument does not capture. When the expected number of detections falls, characterization time is freed and the survey reallocates; the yield is therefore a sublinear function of the occurrence rate, and a counterfactual about the planet population cannot be answered by rescaling a published number.

This has a practical consequence for anyone outside the AYO development team who wants to ask such a counterfactual: they must reimplement the model. That is what we did. Our target question was a mass-selected sample in a narrowed habitable zone (Sec. 6), but the reimplementation is the substantive part of the work, and its result is of wider interest than the counterfactual that motivated it, because it is a test of whether the published description of AYO is sufficient to reconstruct AYO. Our finding is that it very nearly is. The model we built from the papers agrees with the pub-

lished record on 22 of 24 independent comparisons, reproduces the shape of the published yield distribution, and lands within 7.0% of the headline 6 m yield with no parameter fitted to it. That level of agreement is not the interesting part of this paper. The interesting part is the residue: two disagreements that we have been unable to remove, that are not independent of each other in their consequences, and that between them control the quantities a mission designer would care most about — the target mix, the characterization time budget, and the width of the yield distribution.

We argue that neither residue is a bug in our implementation. Each traces to a point where the published description admits more than one reading, and in each case we can show that the readings are distinguishable, that the published numbers do not settle which is intended, and that the choice materially changes the answer. Sections 4 and 5 present the evidence for that claim. Section 7 states the two questions and argues that the reproduction hinges on them.

We have deliberately resisted the temptation to close either gap by tuning. An earlier version of this pipeline carried a throughput calibration factor κ , fitted to the published 6 m yield. It has been removed, for reasons set out in Sec. 2.4 that are themselves an argument about reproducibility: a free multiplier fitted to the quantity you are trying to reproduce does not merely inflate your agreement, it conceals every other error whose signature is a near-uniform change in exposure time.

2 An independent implementation of AYO

2.1 Scope and sources

We reimplemented the detection and characterization model of Stark et al. (2019) and the survey construction of Stark et al. (2014), with the mission parameters, coronagraph performance and astrophysical distributions of Stark et al. (2024). Nothing is taken from the AYO codebase, which is not public; where a quantity is published only as a figure we digitized the vector content stream of the e-print rather than reading values by eye, a decision forced on us early when six values read off a coronagraph figure proved wrong by up to a factor of three.

The target sample is the real HPIC (Tuchow et al. 2024). The coronagraph is the DMVC6 design, with core throughput and raw contrast digitized from the published curves. Exozodi is drawn from the LBTI HOSTS maximum-likelihood distribution digitized from Stark et al. (2024) Fig. 9, with four stars pinned to their measured values. The occurrence prior follows SAG13, normalized to the $\eta_{\oplus}$ of Stark et al. (2024) Sec. 3.6 — a choice that becomes the subject of Sec. 5.

2.2 The model as a graphical model

Writing the pipeline down as a probabilistic graphical model (Fig. 1, Table 1) is not decoration. It makes one distinction precise that the yield literature often leaves implicit: which uncertainties reach the survey’s decision node, and which do not. An uncertainty that reaches the allocation is *actionable* — knowing it before launch would change what the survey does — and one that does not is merely a spread on the outcome. $\eta_{\oplus}$ and the exozodi levels z_i reach the allocation; the per-planet albedo A_{ij} and the per-star planet count N_i do not. That distinction is what makes the counterfactual of Sec. 6 a re-optimization rather than a rescaling.

2.3 The nodes

2.4 What we do not calibrate

Earlier versions of this pipeline carried a scalar throughput factor κ applied to every band, fitted so that the canonical box reproduced the published 22.5 candidates. It reached 1.62 at one stage. It has been removed, and the pipeline now adopts $\kappa = 1.000$, with the fitted value retained only as a diagnostic.

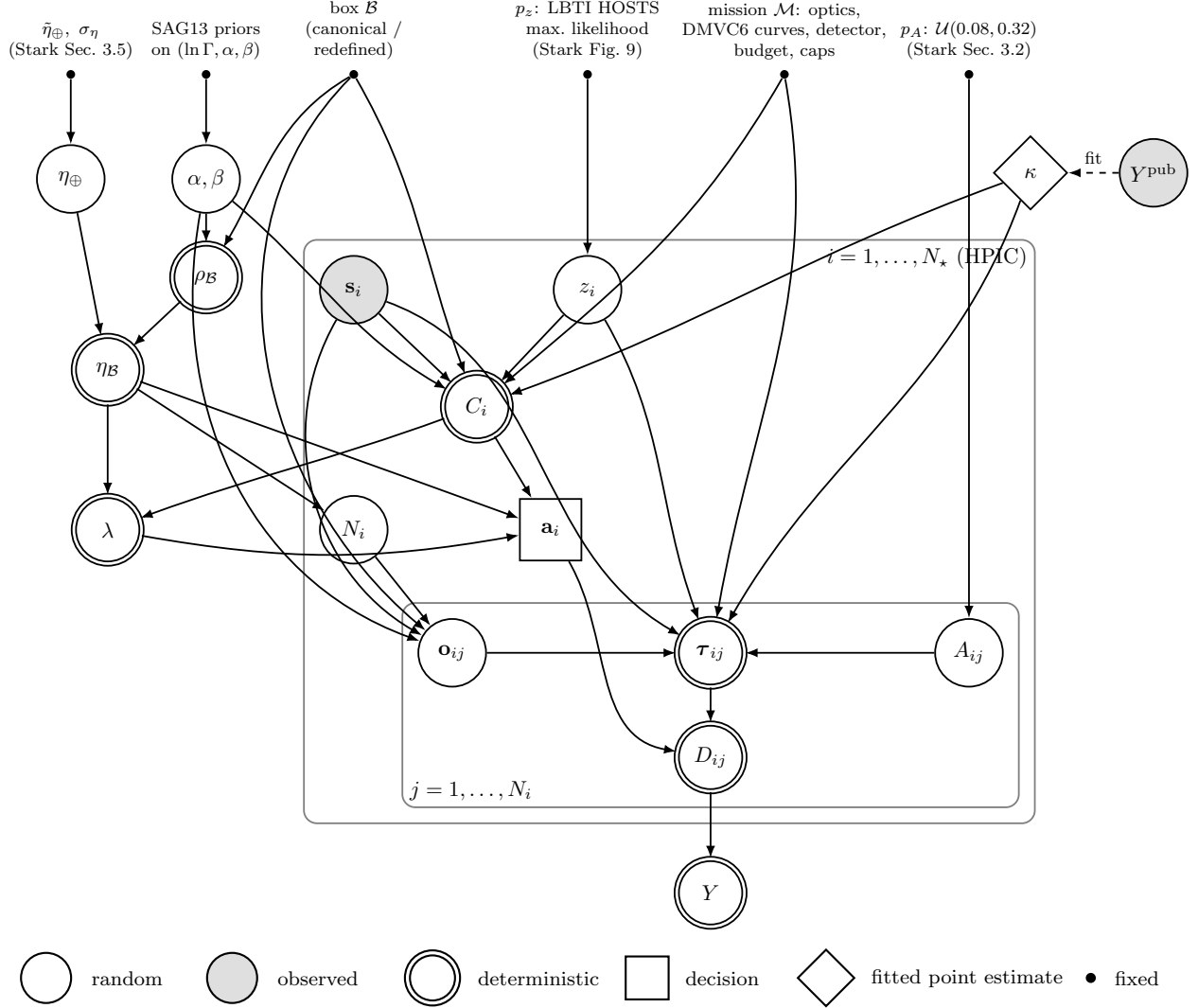


Figure 1: The yield model as an influence diagram. Outer plate: the $N_{\star}$ screened HPIC stars; inner plate: the N_i planets actually present around star i . The survey allocation $\mathbf{a}_i = (\tau_i, k_i)$ — exposure per visit and number of visits — is the one decision node; it is set by the equal-slope multiplier λ from the planner’s completeness curves C_i and the box occurrence rate $\eta_{\mathcal{B}}$. A planet is counted ($D_{ij} = 1$) when its detection time at some visit is within the allocation and its spectral characterization time is within the two-month cap; $Y = \sum_{ij} D_{ij}$. The throughput factor κ is not a random variable: it is fitted once, at fixed exozodi, to the published expected yield $Y^{\text{pub}} = 22.5$ (dashed), and comes out at 1.000, so the edge carries almost no information. Four stars with LBTI-measured exozodi have z_i observed rather than drawn (not shown). The edges that decide the uncertainty budget are the ones *into* $\mathbf{a}_i$: $\eta_{\oplus}$ and z_i reach it (actionable), A_{ij} and N_i do not (static).

Node	Type	Specification	Source; step
$\eta_{\oplus}$	random, global	lognormal, median 0.257, ln-width 0.76 (Sec. 5)	Stark+2024 Sec. 3.5, read at $z = 1$; A06
α, β	random, global	SAG13 shape; posterior of an <code>emcee</code> chain, used only through $\rho_{\mathcal{B}}$	SAG13; A06
$\rho_{\mathcal{B}}$	deterministic	$\eta_{\mathcal{B}}/\eta_{\oplus}$, the SAG13 integral over $\mathcal{B}$ relative to the canonical box; independent of Γ	A06
$\eta_{\mathcal{B}}$	deterministic	$\eta_{\oplus} \rho_{\mathcal{B}}$	A06
$\mathbf{s}_i$	observed	$d, L_{\star}, T_{\text{eff}}, R_{\star}$, HIP id	HPIC (Tuchow+2024); A01
z_i	random, per star	exozodi level, $z_i \sim p_z$; four stars fixed at LBTI values; 200 joint draws of $\mathbf{z}$	Stark+2024 Sec. 3.3, Fig. 9; A02, A04
C_i	deterministic	planner’s completeness $C_i(\tau, k)$ at $A_G = 0.2$, over exposure and visits, counting only planets that are also characterizable	Stark+2019; A04
λ	deterministic	equal-slope multiplier closing the two-year budget	Stark+2014; A05
$\mathbf{a}_i$	decision	(τ_i, k_i) : the vertex of star i ’s concave cost envelope at slope λ	A05, A07
N_i	random, per star	$N_i \sim \text{Poisson}(\eta_{\mathcal{B}})$	Stark+2024 Sec. 3.1; A07
$\mathbf{o}_{ij}$	random, per planet	circular orbit: a, R from SAG13 over $\mathcal{B}$; $\cos i \sim \mathcal{U}$; independent epoch per visit	ExoVista/AYO; A04
A_{ij}	random, per planet	geometric albedo, $\mathcal{U}(0.08, 0.32)$	Stark+2024 Sec. 3.2; A04
τ_{ij}	deterministic	detection time per visit (two channels in quadrature, S/N 7) and characterization time (R 140, best of nine bandpasses, best orbital phase)	Stark+2019 Eqs. 1–9; A04
D_{ij}	deterministic	$\mathbb{K}[\min_{v \leq k_i} \tau_{ijv}^{\text{det}} \leq \tau_i \wedge \tau_{ij}^{\text{char}} \leq 2 \text{ mo}]$	Stark+2019 Sec. 3; A04
Y	deterministic	$\sum_{i,j} D_{ij}$	A07
κ	fitted	throughput factor on every band, fitted to $Y^{\text{pub}}: 1.000$	A03

Table 1: Every node of Fig. 1, what it is, and where it comes from.

Two things forced that decision. The first is that no Stark paper contains such a factor: 22.5 is an output of AYO, not an input to it, so a reproduction that fits to it is not reproducing the same quantity. The second is more serious. We measured the effect of every lever we had tested on the per-planet exposure times, and found that at refitted κ each of them changes those times by a nearly uniform factor of 1.0 ± 0.05 . A calibration with elasticity ~ 0.4 in throughput can absorb any such lever completely. The factor was therefore not a harmless convenience; it was a mechanism that made a whole class of modelling errors invisible, and its removal is what allowed the two residues of Secs. 4 and 5 to become visible as residues rather than being silently rescaled away.

Most of what κ had been absorbing turned out to be one missing term. Stark et al. (2019) Eqs. 5–6 multiply the zodiacal and exozodiacal backgrounds by an extended-source throughput $T_{\text{sky}}(x, y)$, the PSF convolved with a uniform background, which the point-source terms do not carry. Our model had used its large-separation value everywhere. Reconstructing it as a function of separation raises the uncalibrated yield to 20.92 against the published 22.5. We flag that this reconstruction is ours: the term is sourced, its radial shape is not published, and our shape is known to differ from AYO’s in the inner working angles.

3 Agreement with the published record

We hold ourselves to comparisons against numbers stated in the published *text*, and we tag as “tuned” any comparison against a quantity that entered our model as an input, so that it can never be counted as evidence. On that basis 22 of 24 comparisons agree (Appendix A). Nine of Stark’s figures are reproduced in their own axes alongside the originals (Appendix B).

Three agreements are worth singling out, because they are the ones that constrain the model most and because they frame the disagreements that follow.

The yield level. The uncalibrated 6 m expected yield is 20.92 against a published 22.5, a difference of 7.0% with nothing fitted.

The realized-yield distribution. Marginalizing over the occurrence posterior and Poisson counting reproduces the published Fig. 10 closely: mean 21.0 against 21.0, median 18 against 18, $P(Y < 12) = 0.28$ against 0.29, and peak 11 against 10. We stress that the peak is a fragile statistic and should be computed as the exact Poisson mixture over the sampled expected yields rather than as the mode of a histogram of realized draws; the latter estimator, bootstrapped, cannot distinguish 10 from 16.

An independent implementation. EXOSIMS (Savransky & Garrett 2016), which shares no code and no scheduling method with either AYO or this work, agrees with our radiometry on identical stars and optics: local zodi 1.001, exozodi 0.975, stellar photon rate 0.979. EXOSIMS also *found* two of our errors, by carrying terms we lacked, each then confirmed against Stark’s own papers rather than against EXOSIMS.

Against that background, the two sections that follow concern what did not come out right.

4 First discrepancy: the characterization time budget

4.1 The symptom

Our spectral characterization times are too short for easy targets and too long for hard ones. Averaged over the eighteen highest-priority candidates, we require 14.3 d at 6 m and 2.05 d at 9 m, against published values of 22 and 3.5 d. The ratio between apertures is therefore 7.0 against a published 6.2: our model’s characterization cost falls off far more steeply with aperture than Stark’s.

The same steepness appears in the target mix. Against Stark et al. (2024) Fig. 11 our survey allocates almost nothing to roughly half of the targets beyond 15 pc and above $2 L_{\odot}$. The mechanism is not mysterious in itself: at $R = 140$ near $1 \mu\text{m}$ the background is exozodi-dominated, the exozodi count rate is close to distance-independent, and the required time therefore scales as d^4 . An Earth twin needs 0.2 d at τ Cet and 255 d at 110 Her, against a two-month cap. What is mysterious is how AYO’s own survey reaches the completeness its figure shows for those stars, because AYO’s own exposure-time calculator gives comparable times: the Stark et al. (2025) ETC benchmark yields Earth-twin H_2O spectra of 37 d at 12.3 pc and 104 d at 18.2 pc, for a slightly better telescope than the baseline.

We have verified that our implementation of the exposure-time equations reproduces AYO’s own per-observation times to 0.94–1.02 when given AYO’s own inputs. The equations are not where we differ.

4.2 The throughput family cannot supply the difference

Before concluding that the discrepancy is irreducible we tested whether it could be absorbed by the parameters that plausibly could absorb it: a throughput calibration on all bands, a separate calibration on the spectral bands only, the normalization of the coronagraph leak term, and the radial shape of T_{sky} . These are the four levers that a reader might reasonably suspect we had set wrongly, and they are exactly the family that κ used to hide.

We evaluated a 56-point Latin hypercube over that four-parameter box, recording at each point the 6 m planning yield, the characterization times at 6 and 9 m, and the per-distance target completeness against Fig. 11 (Fig. 2). The result is negative and it is clean:

- Over all 56 directly evaluated points, the first-eighteen characterization time never exceeds 16.9 d, against the published 22 d. No point in the box reaches it.
- Restricting to the 20 points that reproduce the published 6 m yield to within 10%, the characterization time spans only 7.0–11.6 d at 6 m and 0.79–1.49 d at 9 m. There is no overlap with the published values at either aperture.

- The posterior constrained on the yield alone puts $\kappa = 1.43$, $\kappa_c = 1.08$ with a correlation of -0.79 between their logarithms, and forces the 6 m characterization time to 9.8 d (6.4–12.7).
- The singular values of the local sensitivity are 31.72, 2.31, 1.50, 0.90: a ratio of 13.7 between the first and second, meaning these six published observables constrain essentially one combination of the four parameters, and that combination is the κ – κ_c anticorrelation.

Characterization time *can* be bought inside this family, but only by surrendering the yield: the posterior constrained on every published observable drives κ_c onto its prior floor trying to slow spectra down, and the yield falls to 16.4. Buying agreement that way is not a reproduction.

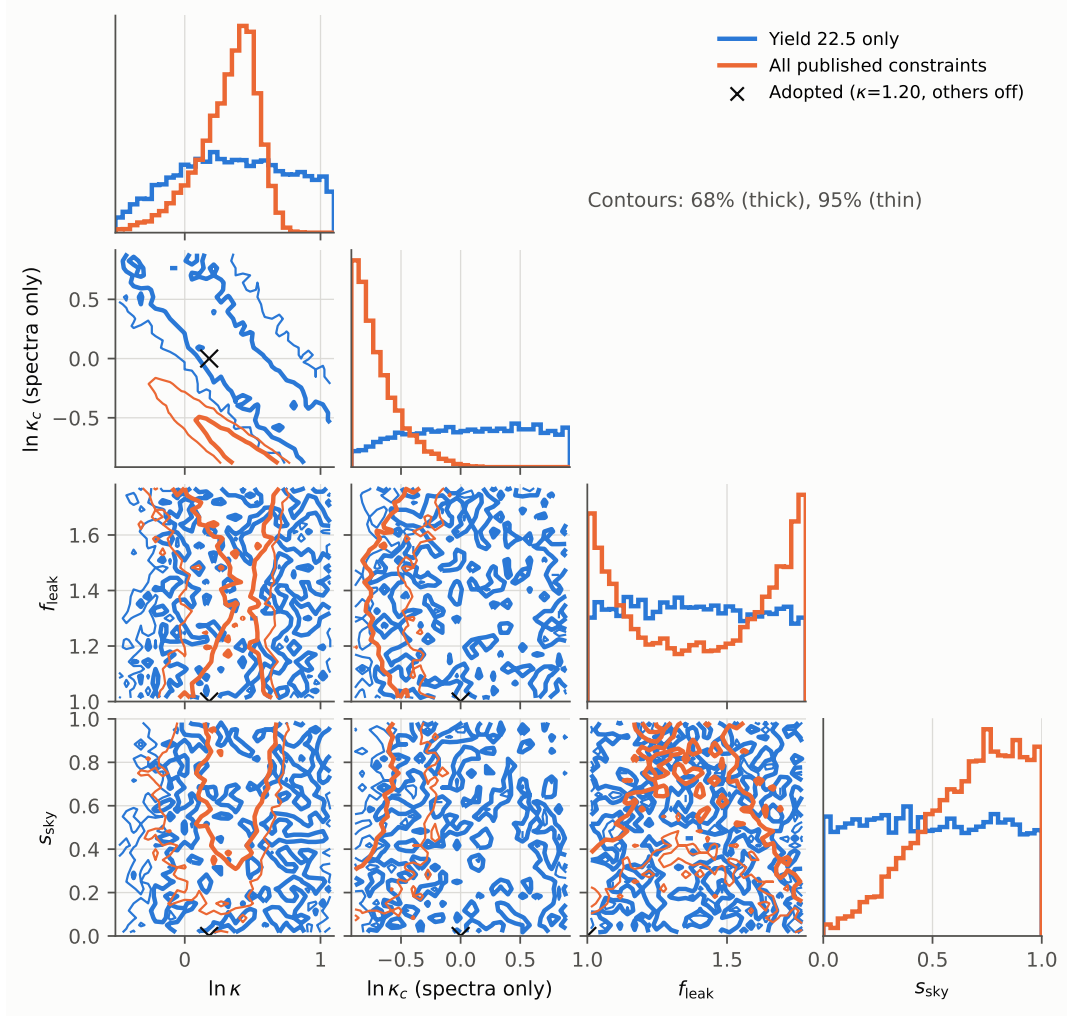


Figure 2: Posteriors over the four throughput-family parameters, from a quadratic emulator of the 56-point design whose leave-one-out error is folded into the tolerance so that emulator error cannot pose as a constraint. Blue: constrained on the published 6 m yield alone. Orange: constrained on the yield, the characterization times at 6 and 9 m, and the Fig. 11 completeness in three distance bins. The $\ln \kappa$ – $\ln \kappa_c$ panel shows the degeneracy that the published observables leave open; the orange contours run into the prior boundary on κ_c while still failing to reach the published characterization times.

4.3 What this implies

We can state the disagreement sharply, which is what makes it a question rather than a complaint. Our exposure-time equations reproduce AYO’s. Our count rates reproduce AYO’s. AYO’s own published ETC gives Earth-twin spectra at 12–18 pc that take 37–104 d. The two-month cap is stated. And yet Stark et al. (2024) Fig. 11 shows those stars reaching habitable-zone completeness

of 0.5–0.8. We cannot construct a survey in which both are true, and we have shown that no setting of the parameters that could plausibly absorb the difference makes them both true.

The resolution is therefore something about how AYO counts or charges characterization within the survey that is not in the papers we have. That is Question 1 (Sec. 7).

5 Second discrepancy: the reading of the occurrence interval

5.1 One sentence, two readings

Stark et al. (2024) Sec. 3.6 states:

“Performing this computation for each element of our posterior, we get an $\eta_{\oplus}$ with mean and 86% confidence interval $\eta_{\oplus} = 0.26^{+0.29}_{-0.14}$.”

The underlying distribution is built by uniformly mixing the two completeness-extrapolation cases of Bryson et al. (2021), and is described as having a tail of large $\eta_{\oplus}$ values.

Propagating this requires two interpretive decisions that the sentence does not settle for a reader building a parametric law. Is 0.26 the mean, as stated, or the centre of the quoted interval? And is [0.12, 0.55] an 86% interval, as stated, or the 68% that is the near-universal convention and a digit transposition away? Our pipeline reads the interval at 68%, giving a lognormal ln-width of 0.76, and anchors 0.26 at the geometric centre — that is, as a median. Both of those depart from the text.

We flag this prominently because it is the kind of choice that reproductions usually bury. It was adopted because it reproduces the published Fig. 10, and a choice adopted because it reproduces the target is exactly the kind of choice that has to be justified on other grounds or declared.

5.2 A factorial decomposition

Our pipeline differs from a fully text-faithful reading in two settings: the albedo method (we recompute detection at the drawn albedo; Stark et al. 2024 Sec. 3.2 describes a per-visit flux test) and the interval reading above. Those two settings form a complete 2×2 , which we evaluated end to end (Table 2).

The decomposition is unambiguous. On the total-variation distance to the published Fig. 10, the interval reading is worth +0.0936, the albedo method -0.0038, and the interaction +0.0007. **The interval reading accounts for 96.1% of the disagreement and the albedo method 3.9%** — and the albedo effect runs in the direction that *favours* Stark’s own described method. Our departure from the published text on the albedo method is therefore not doing useful work, and should be abandoned; we retain it here only so that the factorial is complete, and note the cost, which is that our albedo penalty then falls to 3.5% against a published 12%.

5.3 Following every documented choice makes the agreement worse

The natural response to a reproduction that departs from its source in two places is to demand that it stop departing. We did that, and it fails (Fig. 3). Adopting the per-visit albedo test *and* the 86% interval together moves the peak from 11 to 15 against a published 10, and the total-variation distance from 0.038 to 0.128.

This is the central empirical fact of this section, and we do not think it is a comfortable one for either side. Read one way it says our model is wrong in a manner that only the 68% reading happens to compensate. Read the other way it says the published text and the published figure are not mutually consistent, and that the figure is the better guide to what was computed. We present

Model	Peak	Mean	$P(Y < 12)$	TV to Fig. 10	Yield ratio
Adopted	11	21.0	0.28	0.038	0.2386
Albedo: Stark per-visit test	12	22.5	0.26	0.034	0.2316
$\eta_{\oplus}$ interval as 86%	15	19.5	0.22	0.132	0.2363
Constant T_{sky}	9	17.9	0.37	0.113	0.2299
Every documented choice at once	15	20.6	0.19	0.128	0.2293
All three, incl. constant T_{sky}	12	17.5	0.29	0.149	0.2201
Stark’s own $\eta_{\oplus}$ law	7	16.7	0.44	0.173	0.2344
Stark et al. (2024)	10	21.0	0.29	—	—

Table 2: Seven models, each differing from the adopted one in named settings, evaluated end to end with every other setting, input file and random seed held fixed and κ pinned at 1.000. “Peak” is the mode of the exact Poisson mixture. The last column is the deliverable of Sec. 6; it moves by less than 8% across the whole table.

evidence below for the second reading, but we want to be explicit that the first cannot be excluded by anything we have.

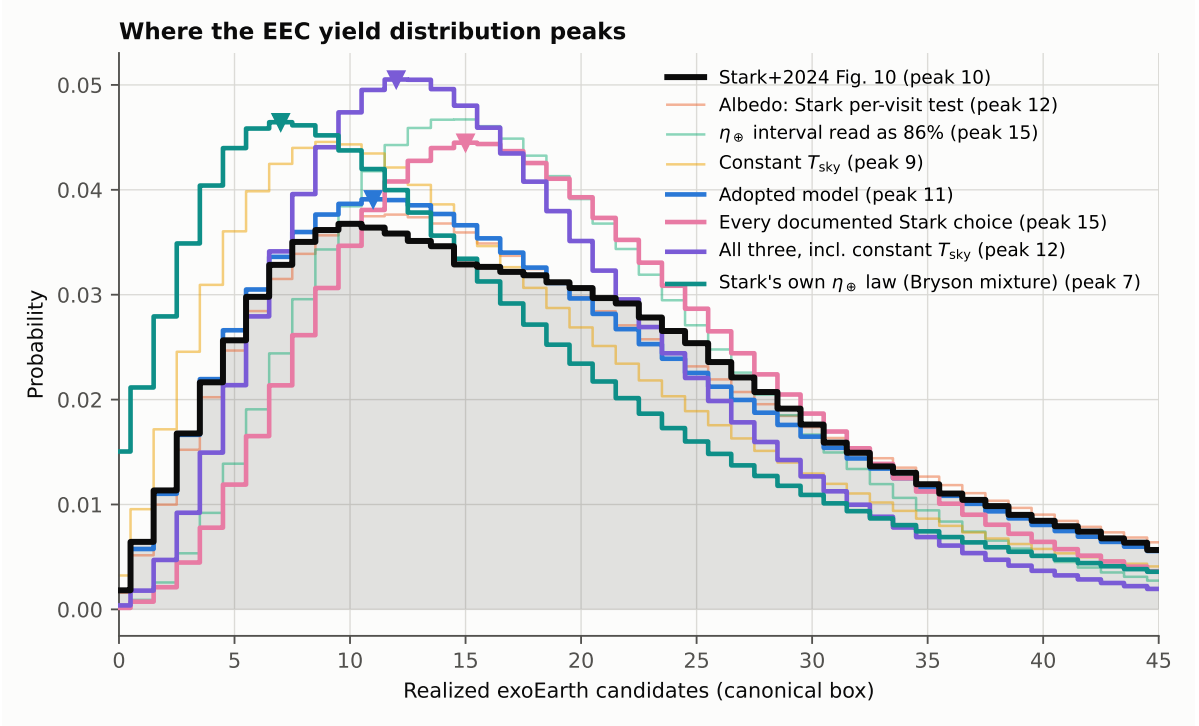


Figure 3: Realized EEC yield distributions for every model of Table 2, against the digitized Stark et al. (2024) Fig. 10 (heavy black). Each curve is the exact Poisson mixture over that model’s sampled expected yields. The one-at-a-time probes are drawn back; the combined models carry peak markers. The adopted model tracks the published curve across the whole range; every model that adopts the 86% interval is displaced to the right and is too sharply peaked.

5.4 Testing the interval against its own source

The reading can be tested without reference to our yield model at all, by asking what the quoted interval means for the distribution it is quoted from. We took the two Bryson et al. (2021) cases, mixed them uniformly as Stark et al. (2024) describes, and rescaled the mixture by the single factor that sets its mean to the stated 0.26. No parameter is fitted.

That distribution puts 59% of its mass inside the quoted $[0.12, 0.55]$. Its own 68% interval is $[0.076, 0.429]$ and its own 86% interval is $[0.047, 0.653]$ — the latter far wider than what is printed.

Its median is 0.183 when its mean is 0.26, which is strongly right-skewed and therefore inconsistent with reading 0.26 as a median, as our pipeline does.

Two conclusions follow. The quoted interval is much closer to a 68% interval of its own source than to an 86% one, which supports the transposition reading. But the mixture is also not a lognormal, and its shape cannot be represented by one: no lognormal can simultaneously carry the stated mean, the stated interval and any single confidence level. Our 68% reading may therefore not be a corrected typo at all — it may be a width that compensates for a tail our functional form does not have. Those two explanations are observationally identical in the figure agreement, which is precisely why the agreement alone never justified the choice.

We tested the obvious remedy: replacing the lognormal with Stark’s own construction, the rescaled Bryson mixture, as the occurrence law. It does not work either. That model peaks at 7 against a published 10 and has the largest total-variation distance of any model we have built (0.173). We believe this reflects our shape transfer rather than his method: Bryson’s published distribution is for his own radius and effective-temperature selection, while Stark recomputed $\eta_{\oplus}$ from Bryson’s chains over a different one, and rescaling transfers the first moment but not the shape. The three candidate laws bracket the published curve — the Bryson mixture at 7, the 68% lognormal at 11, the 86% lognormal at 15, against 10 — and we cannot choose among them from published information. That is Question 2 (Sec. 7).

6 The counterfactual, and its robustness

We now answer the question that motivated the reimplementaion: what the baseline HWO mission would find if exoEarth candidates were defined by mass, $0.5\text{--}2\,M_{\oplus}$ with $R = M^{0.27}$, in a habitable zone of $0.96\text{--}1.20$ AU scaled as $\sqrt{L_{\star}}$.

The occurrence rate over the redefined box is 0.0520 against 0.255 over the canonical one, a ratio of 0.204. The median realized canonical yield is 17.8 (90% interval 5.3–49.2) and the median redefined yield is 4.2 (1.2–14.3). The yield ratio is

$$\frac{Y_{\text{redefined}}}{Y_{\text{canonical}}} = 0.239^{+0.030}_{-0.023}.$$

The probability of reaching 25 candidates falls from 32.2% to 1.04%.

The yield ratio exceeds the occurrence ratio, which is at first sight surprising and is real. Two effects contribute: fewer expected detections free characterization time, which the optimizer reallocates; and truncating the outer habitable zone preferentially removes the faintest, most expensive planets, so the candidates that survive are cheaper per detection.

The result is robust to both open questions. This is the reason we are willing to report it while Secs. 4 and 5 remain open. Across the seven models of Table 2 — which span both readings of the occurrence interval, both albedo methods, the presence and absence of the T_{sky} term, and a completely different functional family for the occurrence law — the yield ratio spans 0.2201 to 0.2386, a range of under 8%, comfortably inside the 90% credible interval of any one of them. The ratio is a comparison between two selections evaluated in the same model, and the systematics that trouble the absolute yield largely divide out of it.

7 Discussion: the two questions

We have described a reproduction that succeeds on most of the published record and fails on two points. We now argue that those two points are not a list of remaining work but the hinge of the whole exercise, and that each reduces to a question of fact that one email can settle.

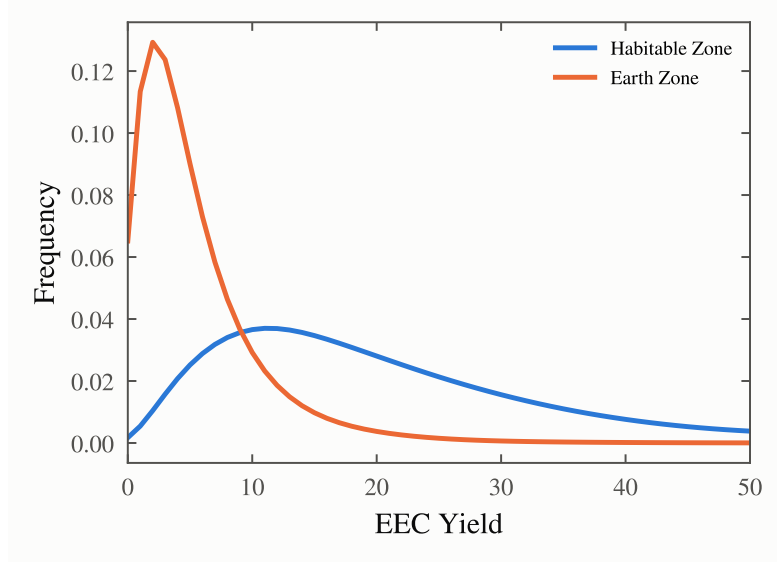


Figure 4: EEC yield distributions for the canonical habitable-zone selection and for the narrowed, mass-selected Earth zone, marginalized over the occurrence posterior and the exozodi draws and including exoplanet sampling uncertainty. Each curve is the exact Poisson mixture over the sampled expected yields. Axes follow Stark et al. (2024) Fig. 10.

Question 1. How does the AYO baseline survey characterize its 10–20 pc targets within the two-month limit?

Stark et al. (2024) Fig. 11 shows habitable-zone completeness of 0.5–0.8 for targets at 10–20 pc. AYO’s own exposure-time calculator (Stark et al. 2025) gives Earth-twin spectra at those distances of 37–104 d. The characterization limit is two months. We reproduce the ETC to 0.94–1.02 on AYO’s own inputs, and we have shown (Sec. 4) that no combination of the throughput parameters that could absorb the difference does absorb it. Either those stars are characterized under conditions more favourable than an Earth twin at maximum elongation, or the completeness plotted in Fig. 11 is not gated on characterization, or characterization is charged to the budget differently from the way we charge it. Any of the three would be a one-sentence answer, and each implies a different correction.

Question 2. In Sec. 3.6, is $\eta_{\oplus} = 0.26$ the mean or the median, and is $[0.12, 0.55]$ an 86% or a 68% interval?

The text says mean and 86%. Our pipeline uses median and 68%, and that combination reproduces the published Fig. 10 while the stated combination does not (Sec. 5). The quoted interval contains 59% of the mass of its own source distribution, which is much nearer 68% than 86%. “86” is a digit transposition of “68”, and the phrase occurs exactly once in the paper, which otherwise quotes standard deviations throughout. We are not in a position to assert a typographical error in someone else’s paper on that evidence, and we do not. But the question is answerable immediately by the authors, and nothing else we can do resolves it.

Why the reproduction hinges on these

It would be possible to read the preceding sections as a report of two modest residuals in an otherwise successful reproduction. We think that reading is wrong, for three reasons.

First, they control the quantities that matter for mission design. The characterization time budget determines the target mix, and the target mix determines which stars must be surveyed for exozodi

in advance, which drives a precursor-mission argument. The occurrence interval controls the width of the yield distribution, and the width determines $P(Y \geq 25)$, which is the quantity the decadal recommendation is written in. A reproduction that matches the mean yield but not the width does not reproduce the statement the mission is designed against.

Second, the agreement we do have is partly contingent on the choices in question. Our headline agreement with Fig. 10 — mean, median and peak — holds under the 68% reading and fails under the 86% one. We cannot simultaneously claim that our model is validated by that agreement and that the reading which produces it is an open question. Honesty requires reporting the agreement as conditional, and Question 2 is the condition.

Third, the two are not independent in their consequences. Both act on the same downstream quantity. The characterization-time discrepancy moves the target mix and hence the level; the occurrence reading moves the width. A reader trying to judge whether our model is fit for a given purpose needs both answered, because a model with the right level and the wrong width and a model with the wrong level and the right width fail different applications.

We therefore present this work not as a completed reproduction with caveats, but as a reproduction that has been carried as far as the published record permits, together with the two specific facts that would carry it the rest of the way.

What we are not claiming

We are not claiming an error in AYO. Every discrepancy reported here is equally consistent with an error in our reimplementation that we have failed to find, and we have found and fixed seventeen such errors already (Appendix C); there is no reason to believe the count is final. What we do claim is narrower and, we think, defensible: that the published description is not sufficient to determine these two quantities, and that we have exhausted the interpretations available to us.

8 Conclusions

1. An independent reimplementation of AYO, built from the published equations and parameters with no parameter fitted to the published yield, reproduces the Stark et al. (2024) baseline 6 m exoEarth candidate yield to 7.0% and agrees with 22 of 24 independent published comparisons.
2. The realized-yield distribution reproduces the published mean, median and peak. The peak should be computed as an exact Poisson mixture; the mode of a histogram of realized draws is too noisy to support the comparison.
3. Spectral characterization times remain too steep in target difficulty and aperture. A 56-point factorial design establishes that no member of the throughput-parameter family reaches the published characterization times at the published yield.
4. The $\eta_{\oplus}$ interval of Sec. 3.6 admits two readings that differ materially. A 2×2 decomposition attributes 96.1% of the residual disagreement with the published yield distribution to that single choice, and following every documented choice simultaneously makes the agreement worse rather than better.
5. Both residues reduce to questions of fact that the AYO authors can answer in a sentence, and we state them explicitly in Sec. 7.
6. For the counterfactual that motivated the work — a mass-selected sample in a narrowed habitable zone — the yield ratio is $0.239^{+0.030}_{-0.023}$, and it is robust at the 8% level across every model variant spanning both open questions.

A Verification against the published record

Check	Published	Model	Diff.	Tol.	Verdict
eta.Earth over the canonical box (SAG13)	0.24	0.2404	0%	3%	agrees
Uncalibrated 6 m yield	22.5	20.92	7%	20%	agrees
Throughput factor needed to reach 22.5	1	1.195	20%	100%	agrees
Sampling distribution mean (Fig. 4)	22.5	20.91	7%	20%	agrees
Sampling distribution sigma (Fig. 4)	5	4.567	9%	25%	agrees
Albedo penalty on expected yield	0.12	0.08635	28%	50%	agrees
P25 including sigma_eta, 6 m	0.32	0.3215	0%	25%	agrees
Yield-aperture exponent	1.9	1.802	5%	15%	agrees
Exozodi-sampling penalty on expected yield	0.111	0.1217	10%	50%	agrees
P25 excluding sigma_eta, 6 m	0.06	0.03771	37%	50%	agrees
Mean characterization time, first 18 EECs, 6 m (days)	22	14.34	35%	30%	disagrees
Mean characterization time, first 18 EECs, 9 m (days)	3.5	2.053	41%	30%	disagrees
Characterization-time ratio 6 m / 9 m	6.2	6.987	13%	20%	agrees
P25 including sigma_eta at 6 m	0.32	0.3203	0%	25%	agrees
P25 including sigma_eta at 7 m	0.53	0.4885	8%	25%	agrees
P25 including sigma_eta at 8 m	0.67	0.6305	6%	25%	agrees
P25 including sigma_eta at 9 m	0.78	0.7285	7%	25%	agrees
Fixed-eta yield mean (Fig. 10 red)	17.35	16.81	3%	10%	agrees
Realized-yield distribution median	18	18	0%	15%	agrees
Realized-yield fraction below 12 EECs	0.2907	0.2825	3%	35%	agrees
Realized-yield distribution mean	21	21.04	0%	30%	agrees
Realized-yield distribution mode	10	12	20%	25%	agrees
Maximum luminosity among reachable targets	20	12.5	38%	60%	agrees
Maximum distance among reachable targets	25	22.02	12%	60%	agrees

Table 3: The 24 independent comparisons of step A09, of which 22 agree. Published values taken from Fig. 10 are digitized from the figure’s vector content stream. Comparisons against quantities that entered the model as inputs are tagged as tuned and excluded from the count.

B Published figures beside their model counterparts

C Ledger of hypotheses tested

The reproduction proceeded by forming hypotheses about the source of each disagreement and testing them. We record the full ledger because the distinction between a hypothesis rejected on evidence and a real defect that turned out not to explain the symptom is load-bearing: a defect can be genuine, worth fixing, and irrelevant to the symptom that led to it. Four separate times an apparent agreement rested on two errors that cancelled, which is the reason this paper reports conditional agreement rather than agreement.

D Every contributing star

Star	SpT	d (pc)	L ($L_{\odot}$)	Earth zone (mas)	classic HZ (mas)	C_{HZ}	C_{EZ}	N_{HZ}	N_{EZ}	ratio
HD 95735	M2+V	2.55	0.0239	58–73	58–101	0.99	1.00	0.247	0.052	0.21
eps Ind	K5V	3.64	0.222	124–155	123–216	0.98	1.00	0.246	0.052	0.21
tau Cet	G8V	3.65	0.52	189–237	188–330	0.98	1.00	0.246	0.052	0.21
61 Cyg A	K5V	3.50	0.158	109–136	108–190	0.98	1.00	0.245	0.052	0.21
61 Cyg B	K7V	3.50	0.104	89–111	88–154	0.98	1.00	0.244	0.052	0.21
HD 217987	M2V	3.29	0.0413	59–74	59–103	0.97	1.00	0.241	0.052	0.21
70 Oph A	K0V	5.11	0.514	135–168	133–234	0.95	1.00	0.235	0.051	0.22
V* AX Mic	M1V	3.97	0.105	78–98	77–136	0.96	1.00	0.235	0.051	0.22
eps Eri	K2V	3.22	0.342	174–218	173–303	0.96	1.00	0.234	0.051	0.22

Star	SpT	d (pc)	L ($L_{\odot}$)	Earth zone (mas)	classic HZ (mas)	C_{HZ}	C_{EZ}	N_{HZ}	N_{EZ}	ratio
del Equ	F7(V)+G	4.51	0.265	110–137	108–191	0.95	1.00	0.234	0.051	0.22
omi02 Eri	K0V	5.01	0.428	125–157	124–218	0.95	1.00	0.233	0.051	0.22
70 Oph B	K4V	5.10	0.162	76–95	75–132	0.92	0.99	0.225	0.051	0.22
e Eri	G6V	6.04	0.663	129–162	128–225	0.91	0.99	0.220	0.051	0.23
36 Oph	K2V+K1V	5.95	0.334	93–117	92–162	0.90	0.97	0.219	0.050	0.23
sig Dra	K0V	5.76	0.433	110–137	108–191	0.89	0.98	0.218	0.051	0.23
IRAS 20079-3614		6.01	0.26	81–102	81–142	0.88	0.94	0.216	0.048	0.22
HD 88230	K6VeFe-	4.87	0.117	68–84	67–118	0.89	0.97	0.214	0.049	0.23
HD 131977	K4V	5.89	0.223	77–96	76–134	0.87	0.94	0.213	0.048	0.23
36 Oph B	K1V	5.95	0.331	93–116	92–162	0.87	0.96	0.212	0.049	0.23
eta Cas	F9V	5.92	1.27	182–228	181–317	0.88	0.99	0.212	0.050	0.24
ksi Boo	G7Ve+K5	6.75	0.554	106–132	105–184	0.86	0.92	0.210	0.047	0.23
V* GX And	M2V	3.56	0.0239	42–52	41–72	0.86	0.88	0.209	0.045	0.22
del Pav	G8IV	6.10	1.25	176–220	174–306	0.87	0.99	0.208	0.050	0.24
HD 219134	K3V	6.54	0.29	79–99	78–137	0.84	0.89	0.207	0.046	0.22
V* V2215 Oph	K5V	5.95	0.158	64–80	63–111	0.84	0.87	0.204	0.045	0.22
107 Psc	K1V	7.64	0.458	85–106	84–148	0.83	0.87	0.201	0.045	0.22
mu. Cas	G5Vb	7.68	0.479	87–108	86–151	0.79	0.82	0.190	0.042	0.22
61 Vir	G6.5V	8.53	0.841	103–129	102–179	0.78	0.84	0.188	0.043	0.23
bet CVn	G0V	8.47	1.22	125–156	124–218	0.78	0.85	0.187	0.043	0.23
HD 16160	K3V	7.23	0.282	71–88	70–123	0.77	0.80	0.187	0.041	0.22
HD 173739	M3V	3.52	0.016	35–43	34–60	0.76	0.69	0.183	0.036	0.20
HD 156384	K3V+K5V	6.97	0.192	60–76	60–105	0.74	0.77	0.180	0.040	0.22
HD 4628	K2.5V	7.43	0.3	71–88	70–123	0.73	0.75	0.175	0.039	0.22
chi01 Ori	G0V	8.70	1.11	116–145	115–202	0.73	0.79	0.174	0.041	0.23
p Eri A	K2V	8.19	0.345	69–86	68–120	0.72	0.77	0.173	0.040	0.23
V* TW PsA	K4Ve	7.60	0.196	56–70	55–97	0.72	0.78	0.172	0.040	0.24
ksi UMa	F8.5:V+	8.73	0.796	98–123	97–171	0.71	0.77	0.172	0.039	0.23
ksi UMa A	F8.5:V	8.73	1.26	123–154	122–214	0.72	0.80	0.170	0.040	0.24
p Eri B	K2V	8.20	0.305	65–81	64–112	0.71	0.76	0.170	0.039	0.23
ksi Boo B	K5Ve	6.75	0.127	51–63	50–88	0.70	0.74	0.170	0.038	0.23
HD 192310	K2+V	8.81	0.405	69–87	69–121	0.71	0.79	0.169	0.041	0.24
zet Tuc	F9.5V	8.61	1.28	126–158	125–220	0.70	0.76	0.169	0.039	0.23
kap01 Cet	G5V	9.27	0.881	97–122	96–169	0.69	0.76	0.164	0.039	0.24
41 Ara	G9V	8.79	0.461	74–93	73–129	0.69	0.75	0.162	0.038	0.24
bet Com	F9.5V	9.20	1.42	124–155	123–216	0.69	0.76	0.162	0.039	0.24
HD 102365	G2V	9.32	0.849	95–119	94–165	0.68	0.77	0.161	0.039	0.24
HD 131976	M1.5V	5.90	0.0723	44–55	43–76	0.67	0.67	0.161	0.035	0.21
chi Dra	F7V	8.30	2.19	171–214	170–298	0.69	0.83	0.161	0.041	0.26
gam Pav	F9VFe-1	9.26	1.46	125–157	124–218	0.67	0.76	0.157	0.039	0.25
61 UMa	G8V	9.57	0.625	79–99	78–138	0.67	0.76	0.157	0.039	0.25
pi.03 Ori	F6V	8.02	2.85	202–253	200–352	0.68	0.86	0.154	0.042	0.27
mu.01 Her	G5IV	8.33	2.68	188–235	186–328	0.67	0.86	0.153	0.042	0.28
gam Lep	F6V	8.90	2.36	166–207	164–288	0.66	0.79	0.152	0.040	0.26
bet Hyi	G0V	7.48	3.57	243–303	240–422	0.64	0.92	0.148	0.044	0.30
alf Men	G7V	10.21	0.866	87–109	87–152	0.64	0.74	0.148	0.038	0.25
del Tri	G0.5VFe	10.91	1.21	97–121	96–168	0.63	0.80	0.146	0.041	0.28
HD 10780	K0V	10.04	0.536	70–88	69–122	0.60	0.73	0.138	0.037	0.27
HD 32147	K3+V	8.84	0.29	58–73	58–102	0.59	0.67	0.138	0.034	0.25
tet Per	F8V	11.14	2.32	131–164	130–228	0.61	0.82	0.137	0.040	0.29
iot Per	G0V	10.57	2.26	136–171	135–237	0.61	0.78	0.137	0.039	0.28
HD 79210	K7V	6.33	0.0813	43–54	43–75	0.57	0.52	0.136	0.027	0.20
del Eri	K0+IV	9.09	3.29	192–240	190–333	0.59	0.82	0.133	0.040	0.30
12 Oph	K1V	9.89	0.46	66–82	65–114	0.58	0.70	0.133	0.036	0.27
V* AK Lep	K3	8.89	0.279	57–71	56–99	0.56	0.63	0.130	0.032	0.25
HD 36395	M1.5Ve	5.70	0.0594	41–51	41–71	0.53	0.47	0.126	0.024	0.19
alf Cen B	K1V	1.35	0.552	529–661	524–921	0.49	0.65	0.126	0.034	0.27
11 LMi	G8Va	11.23	0.801	77–96	76–133	0.53	0.70	0.120	0.035	0.29
zet Dor	F9VFe-0	11.69	1.48	100–125	99–174	0.53	0.72	0.119	0.036	0.30
lam Ser	G0-V	11.90	2.03	115–144	114–200	0.54	0.73	0.119	0.036	0.31
zet TrA	F9V	12.07	1.34	92–115	91–160	0.51	0.71	0.115	0.036	0.31
20 Crt	K0V	9.56	0.371	61–77	61–106	0.48	0.57	0.113	0.029	0.26
gam Ser	F6V	11.16	2.96	148–185	147–258	0.50	0.71	0.111	0.034	0.31
lam Aur	G1.5IV-	12.56	1.78	102–128	101–178	0.50	0.72	0.111	0.036	0.32
HD 50281	K3.5V	8.74	0.219	51–64	51–89	0.46	0.41	0.106	0.021	0.20
bet Vir	F9V	11.00	3.69	168–209	166–292	0.48	0.74	0.105	0.035	0.33
HD 42581	M1V	5.76	0.0515	38–47	37–66	0.44	0.19	0.104	0.010	0.10
iot Peg	F5V	11.90	3.57	152–191	151–265	0.48	0.75	0.104	0.035	0.34
HD 79211	M0V	6.33	0.0707	40–50	40–70	0.44	0.24	0.104	0.012	0.12
zet02 Ret	G1V	12.04	1.02	80–100	80–140	0.46	0.63	0.103	0.032	0.31
36 UMa	F8V	12.94	1.63	95–118	94–165	0.46	0.66	0.100	0.033	0.33
zet01 Ret	G2.5VHd	12.04	0.795	71–89	70–124	0.44	0.53	0.098	0.027	0.27
HD 17925	K1V	10.36	0.401	59–73	58–102	0.41	0.38	0.092	0.019	0.21
HD 191849	M0V	6.16	0.0616	39–48	38–67	0.38	0.15	0.088	0.008	0.09

Star	SpT	d (pc)	L ($L_{\odot}$)	Earth zone (mas)	classic HZ (mas)	C_{HZ}	C_{EZ}	N_{HZ}	N_{EZ}	ratio
ups And	F9V	13.47	3.38	131–164	130–228	0.41	0.67	0.086	0.031	0.36
54 Psc	K0.5V	11.10	0.529	63–79	62–109	0.38	0.39	0.084	0.019	0.23
gam Vir	F1–F2V	12.02	4.55	170–213	169–296	0.38	0.65	0.080	0.029	0.36
10 Tau	F9IV–V	13.92	3.16	123–153	121–213	0.37	0.61	0.080	0.029	0.36
HD 157881	K7V	7.71	0.123	44–55	43–76	0.35	0.16	0.080	0.008	0.10
HD 103095	K1V_Fe-	9.17	0.235	51–63	50–88	0.34	0.23	0.079	0.012	0.15
47 UMa	G1–VFe-	13.88	1.6	87–109	86–152	0.36	0.50	0.077	0.025	0.32
tau01 Eri	F7V	14.27	2.67	110–137	109–191	0.34	0.53	0.075	0.026	0.34
iot Psc	F7V	13.66	3.39	129–162	128–225	0.35	0.59	0.075	0.028	0.37
alf For	F6V	14.00	4.28	142–177	140–247	0.35	0.60	0.072	0.027	0.37
eta Cas B	K7Ve	5.93	0.0513	37–46	36–64	0.29	0.04	0.071	0.002	0.03
HD 119850	M2V	5.43	0.0397	35–44	35–61	0.30	0.06	0.070	0.003	0.04
HD 104304	G8IV	12.76	0.892	71–89	70–124	0.32	0.34	0.069	0.017	0.24
HD 223778	K3V	10.89	0.399	56–70	55–97	0.30	0.19	0.068	0.009	0.14
alf Com	F5V+F6V	14.32	3	116–145	115–202	0.32	0.51	0.068	0.024	0.36
gam Vir B	F0mF2V	12.74	5.27	173–216	171–301	0.32	0.56	0.067	0.025	0.37
HD 147513	G5V	12.89	0.998	74–93	74–129	0.29	0.35	0.065	0.018	0.27
HD 13445	K1.5V	10.76	0.406	57–71	56–99	0.28	0.20	0.065	0.010	0.15
V* DE Boo	K0.5V	11.63	0.541	61–76	60–106	0.29	0.21	0.064	0.010	0.16
HD 122064	K3V	10.06	0.291	51–64	51–89	0.28	0.12	0.062	0.006	0.10
58 Eri	G2.5IV-	13.23	0.966	71–89	71–124	0.28	0.26	0.062	0.013	0.21
psi Cap	F5V	14.63	3.72	127–158	125–220	0.29	0.48	0.060	0.022	0.36
111 Tau	F8V	14.57	1.78	88–110	87–153	0.27	0.35	0.058	0.016	0.28
85 Peg	G5VbFe-	12.32	0.674	64–80	63–111	0.26	0.21	0.057	0.010	0.18
HD 10307	G1V	14.67	1.89	90–112	89–157	0.26	0.37	0.057	0.018	0.32
26 Dra	F9V+K3V	14.43	1.48	81–101	80–141	0.26	0.31	0.055	0.015	0.28
zet Her	G0IV	10.80	7.26	240–300	237–417	0.26	0.49	0.054	0.020	0.37
chi Her	F8VFe-2	15.90	3.16	107–134	106–187	0.26	0.40	0.054	0.019	0.35
HD 211415	G0V	14.06	1.17	74–92	73–129	0.25	0.23	0.053	0.011	0.21
nu. Phe	F9VFe+0	15.25	2.03	90–112	89–156	0.24	0.30	0.050	0.014	0.29
alf Crv	F1V	14.97	4.34	134–167	132–232	0.23	0.39	0.048	0.017	0.36
b Aql	G7IVHde	14.91	1.7	84–105	83–146	0.23	0.28	0.048	0.013	0.28
HD 84117	F9V	14.95	2.01	91–114	90–159	0.22	0.30	0.048	0.014	0.30
tau Boo	F7IV–V	15.60	3.18	110–137	109–191	0.23	0.33	0.047	0.016	0.33
h Dra	F8V	15.33	2.28	95–118	94–164	0.22	0.29	0.047	0.014	0.30
eta Lep	F2V	14.95	5.66	153–191	151–266	0.22	0.37	0.045	0.016	0.35
sig Boo	F4Vkf2m	15.75	3.22	109–137	108–190	0.21	0.29	0.044	0.014	0.31
alf Cen A	G2V	1.35	1.66	918–1147	908–1596	0.18	0.24	0.044	0.012	0.27
171 Pup	F9V	15.27	1.57	79–99	78–137	0.19	0.19	0.042	0.009	0.22
m Tau	G4V	15.91	2.41	94–117	93–163	0.20	0.27	0.041	0.012	0.30
18 Sco	G2Va	14.13	1.1	71–89	70–124	0.19	0.14	0.040	0.007	0.17
mu. Ara	G3IV–V	15.60	1.9	85–106	84–148	0.20	0.22	0.040	0.011	0.27
ksi Peg	F6V	16.39	4.76	128–160	126–222	0.19	0.30	0.040	0.013	0.33
b Her	F6V	15.76	2.02	87–108	86–151	0.19	0.21	0.039	0.010	0.24
HD 160346	K3–V	10.71	0.32	51–63	50–88	0.17	0.02	0.039	0.001	0.03
20 LMi	G3VaHde	14.92	1.37	75–94	75–131	0.18	0.15	0.038	0.007	0.19
HD 69830	G8:V	12.58	0.609	60–74	59–104	0.18	0.05	0.038	0.002	0.06
HD 176051	F9V+K1V	14.89	1.42	77–96	76–134	0.18	0.16	0.038	0.008	0.21
rho01 Cnc	K0IV–V	12.58	0.635	61–76	60–106	0.17	0.06	0.037	0.003	0.08
tet UMa	F7V	13.53	7.9	199–249	197–347	0.18	0.35	0.036	0.013	0.37
HD 166620	K2V	11.09	0.363	52–65	52–91	0.16	0.02	0.035	0.001	0.03
eta Boo	G0IV	11.34	8.79	251–314	248–437	0.16	0.33	0.034	0.013	0.38
bet Tra	F1V	12.42	8.71	228–285	226–397	0.16	0.30	0.032	0.012	0.36
alf CMi	F5IV–V+	3.50	7.21	737–921	729–1282	0.15	0.30	0.031	0.013	0.41
I Car	F3V	16.28	5.25	135–169	134–235	0.14	0.22	0.029	0.010	0.34
tau01 Hya	F5V	18.24	4.05	106–132	105–184	0.13	0.19	0.026	0.008	0.31
gam CrA A	F8V	17.03	2.47	89–111	88–154	0.13	0.12	0.026	0.005	0.20
HD 172051	G6V	13.02	0.682	61–76	60–106	0.12	0.02	0.026	0.001	0.03
i Boo	G0Vn	12.95	0.639	59–74	59–103	0.12	0.01	0.025	0.000	0.02
9 Pup	G0V	16.67	1.95	80–101	80–140	0.12	0.07	0.024	0.003	0.14
HD 3443	K1V+G	15.53	1.32	71–89	70–123	0.12	0.05	0.023	0.002	0.08
HD 188088	K3VaCN1	14.14	0.803	61–76	60–106	0.11	0.01	0.023	0.000	0.02
nu.02 Lup	G3/5V	14.74	1.04	66–83	66–115	0.11	0.02	0.023	0.001	0.03
pi.01 UMa	G0.5V	14.43	0.972	66–82	65–114	0.11	0.01	0.023	0.001	0.03
phi02 Cet	F7V	15.92	1.79	81–101	80–140	0.11	0.07	0.022	0.003	0.15
del Aql	F1IV–V(	15.37	8.26	179–224	178–312	0.11	0.20	0.021	0.008	0.35
gam CrA	F8V+F8V	17.89	4.8	118–147	116–204	0.11	0.13	0.021	0.006	0.27
HD 115404	K2V	10.98	0.302	48–60	48–84	0.09	0.00	0.020	0.000	0.00
HD 37394	K1	12.26	0.477	54–68	53–94	0.09	0.00	0.019	0.000	0.00
70 Vir	G4Va	18.09	3.1	93–117	93–163	0.10	0.10	0.019	0.004	0.22
g Lup	F3/5V	17.39	3.33	101–126	100–175	0.10	0.09	0.019	0.004	0.21
51 Peg	G2IV	15.51	1.38	73–91	72–126	0.09	0.03	0.019	0.001	0.08
tau06 Eri	F5IV–V	17.78	5.15	123–153	121–213	0.09	0.11	0.018	0.005	0.26
58 Oph	F5V	17.17	2.72	92–115	91–160	0.09	0.06	0.018	0.003	0.15

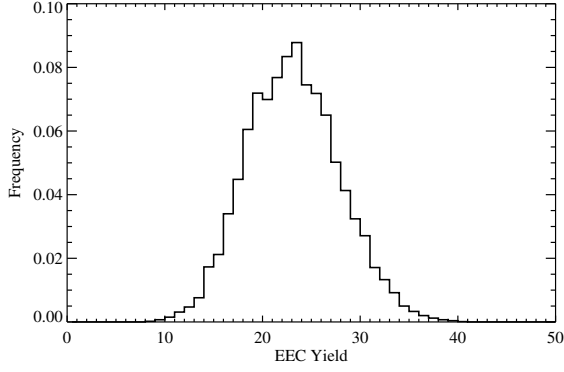
Star	SpT	d (pc)	L ($L_{\odot}$)	Earth zone (mas)	classic HZ (mas)	C_{HZ}	C_{EZ}	N_{HZ}	N_{EZ}	ratio
HD 74576	K3V	11.19	0.318	48–60	48–84	0.08	0.00	0.018	0.000	0.00
ksi Oph	F2V	17.50	4.35	114–143	113–199	0.09	0.09	0.017	0.004	0.22
psi05 Aur	F9V	16.61	1.82	78–97	77–136	0.08	0.02	0.017	0.001	0.06
HD 103932	K4+V	10.17	0.212	43–54	43–76	0.07	0.00	0.016	0.000	0.00
rho Gem	F1V	17.83	5.44	126–157	124–218	0.07	0.08	0.014	0.003	0.22
HD 5015	F9V	18.90	3.62	97–121	96–168	0.07	0.04	0.014	0.002	0.13
HD 217357	K7+Vk	8.23	0.101	37–46	37–64	0.06	0.00	0.013	0.000	0.00
tet Cyg	F3+V	18.42	4.27	108–135	107–187	0.07	0.05	0.013	0.002	0.18
HD 207129	G2V	15.55	1.21	68–85	67–118	0.06	0.00	0.012	0.000	0.01
HD 114837	F6VFe-0	18.23	2.99	91–114	90–158	0.06	0.03	0.012	0.001	0.12
10 UMa	F3V+G5V	18.79	6.73	132–166	131–230	0.06	0.07	0.012	0.003	0.22
eta Crv	F2V	18.24	4.77	115–144	114–200	0.06	0.05	0.011	0.002	0.18
HD 32450	K7V	8.36	0.11	38–48	38–66	0.05	0.00	0.010	0.000	0.00

Stars with an expected yield of at least 0.01 EEC in either zone’s survey, in order of their classic-zone yield. Zone edges are angular separations in milliarcseconds; C is the completeness reached, averaged over exozodi draws; N is the expected number of candidates. The complete table is `TabulateEarthZones/earthZoneTable.csv`.

References

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- [stark2019] Stark, C. C., et al. 2019, *JATIS*, 5, 024009 (arXiv:1904.11988).
- [stark2024] Stark, C. C., et al. 2024, *JATIS* (arXiv:2405.19418).
- [stark2025] Stark, C. C., et al. 2025, arXiv:2502.18556 (ETC benchmark).
- [tuchow2024] Tuchow, N. W., et al. 2024, the HWO Preliminary Input Catalog (HPIC).

Stark et al. (2024) Fig. 4



This model

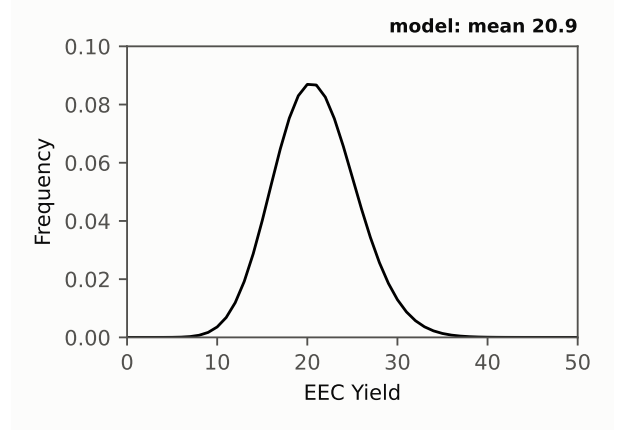
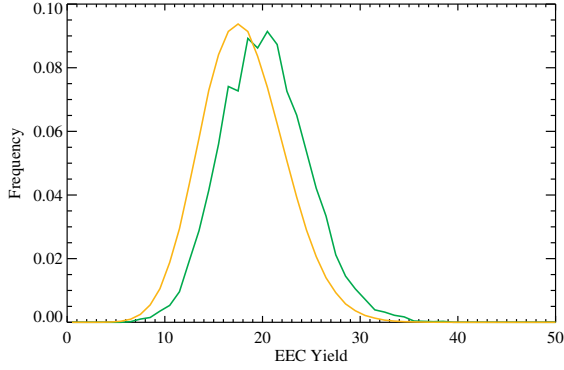


Figure 5: Exoplanet sampling only: $A_G = 0.2$, exozodi fixed at 3 zodis, Poisson counting. Published mean 22.5, standard deviation 21% of the mean. Model mean 20.9, which is *tuned*: κ is fitted to 22.5.

Stark et al. (2024) Fig. 7



This model

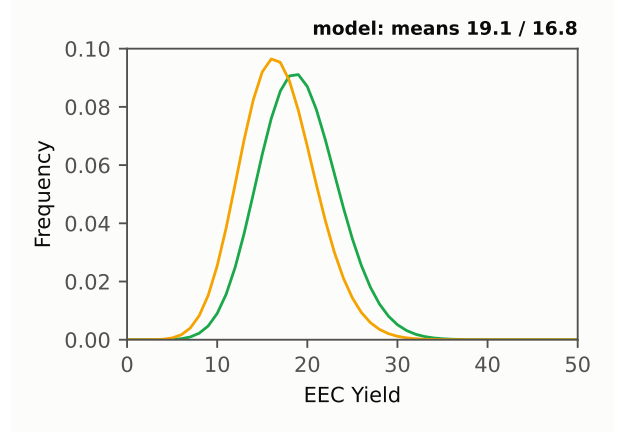
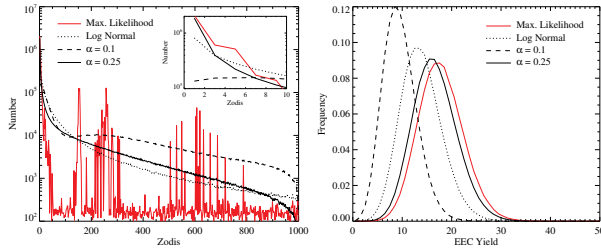


Figure 6: Albedo drawn with exozodi fixed (green), and exozodi drawn as well (orange), $\eta_{\oplus}$ fixed. Published means 19.8 and 17.6. Model means 19.1 and 16.8 over 200 draws.

Stark et al. (2024) Fig. 9



This model

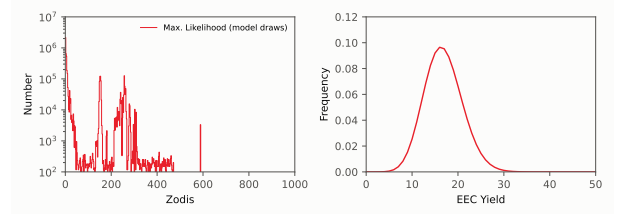
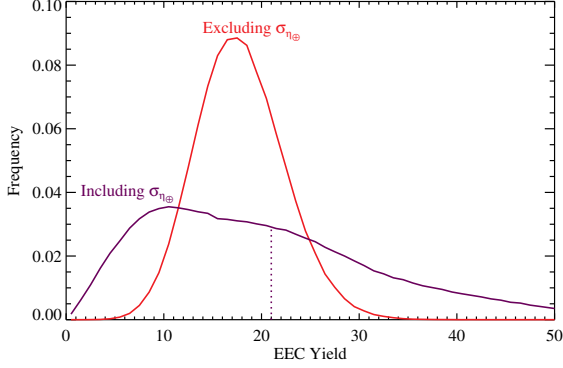


Figure 7: Left: exozodi levels drawn from the LBTI HOSTS maximum-likelihood distribution (red); the model panel is the histogram of the levels the pipeline actually drew, scaled to Stark's 500×10^4 draws (median 3.00 zodis; Stark's other three distributions are not modelled). Right: the $\eta_{\oplus}$ -fixed yield those levels produce (red).

Stark et al. (2024) Fig. 10



This model

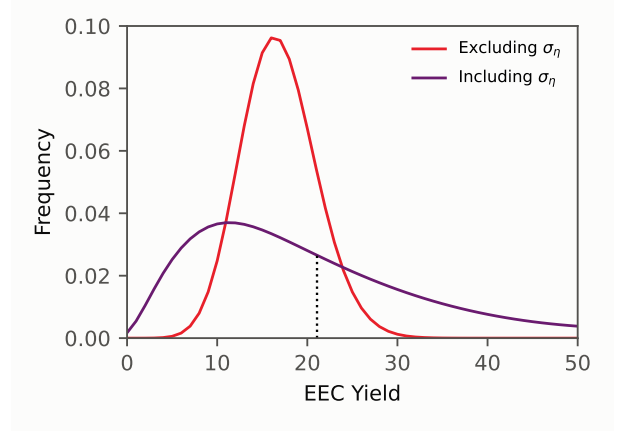
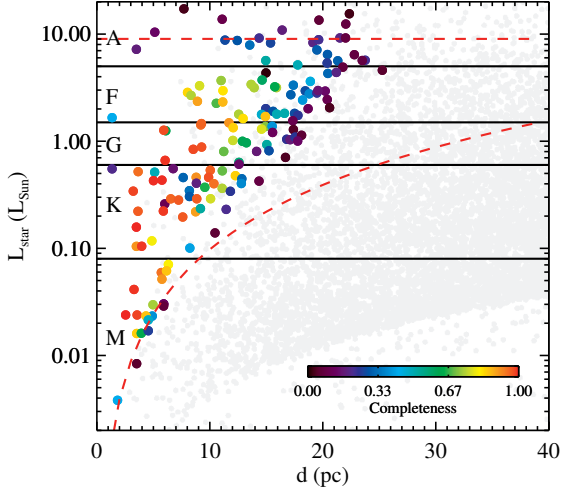


Figure 8: Realized yield including (purple) and excluding (red) $\eta_{\oplus}$ uncertainty; the dotted line marks the purple mean. Total-variation distance, model to published: 0.029 (purple), 0.052 (red). Means: published 21.0 / 17.3, model 21.1 / 16.9.

Stark et al. (2024) Fig. 11



This model

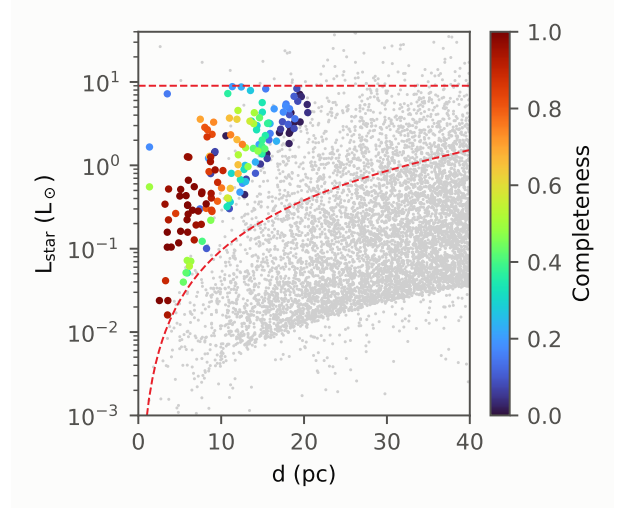
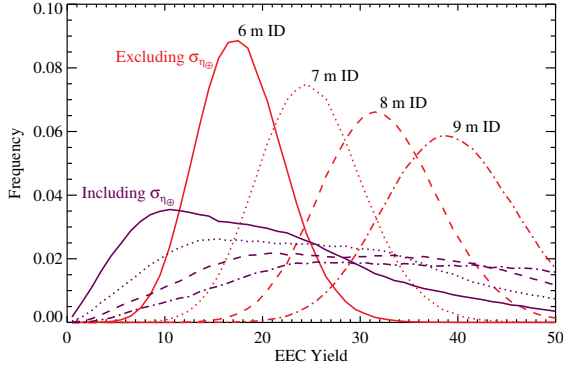


Figure 9: Targets selected in one representative exozodi draw, coloured by completeness, over the input list in grey. Published: 154 targets, summed completeness 77.4, median 0.50. Model (draw 14, the median-yield draw): 157 targets, summed 77.0, median 0.09 over the 141 matched targets. Red dashed lines: the noise floor for a $1.4 R_{\oplus}$ planet at the EEID, and the HZ outer edge at $1.5 \lambda/D$ at $1 \mu\text{m}$; the spectral-type lines are not redrawn.

Stark et al. (2024) Fig. 12



This model

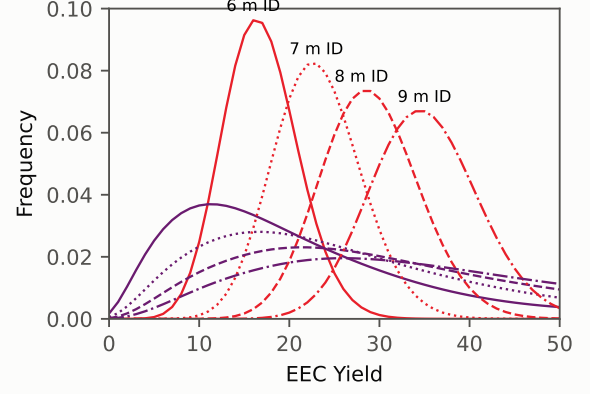
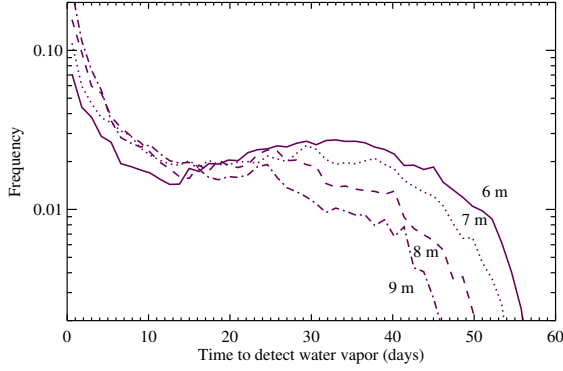


Figure 10: Realized yield at 6, 7, 8 and 9 m inscribed diameter (solid, dotted, dashed, dot-dashed), including (purple) and excluding (red) $\sigma_{\eta_{\oplus}}$, with the calibration frozen at 6 m. Published: excluding $\sigma_{\eta_{\oplus}}$, 9 m gives 2.2 times the 6 m yield; model 2.08 times.

Stark et al. (2024) Fig. 14



This model

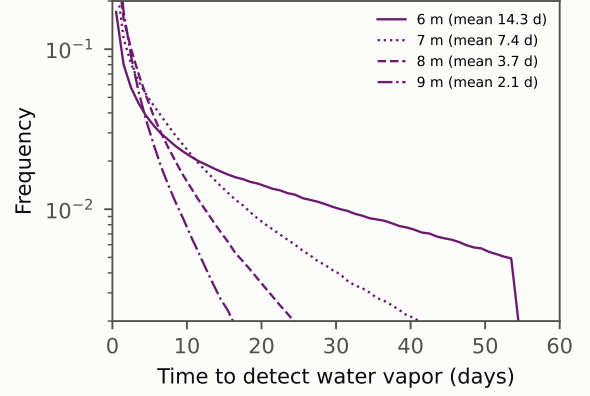
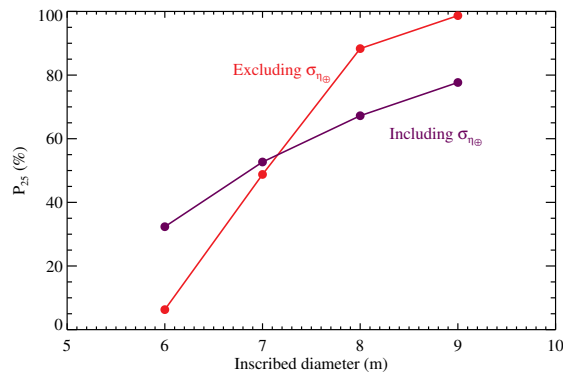


Figure 11: Characterization times of individual EECs among the first 18 at 6–9 m (model: 20 exozodi draws re-derived planet by planet, 1-day bins). Published means 22 and 3.5 days at 6 and 9 m; model means at 6/7/8/9 m: 14.3 / 7.4 / 3.7 / 2.1 days. Stark does not state his normalization, and with 1-day bins his 9 m plateau alone would exceed his stated mean, so compare shapes, not heights: his distributions keep a plateau of 15–50-day spectra at every aperture, which the model's lack.

Stark et al. (2024) Fig. 15



This model

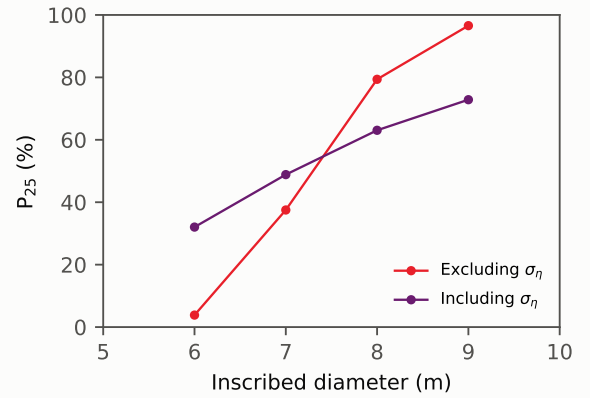
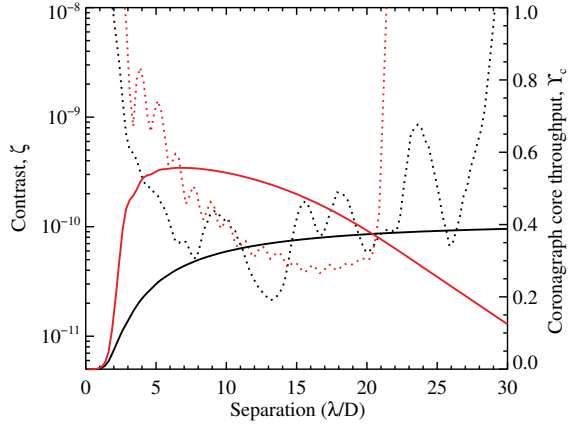


Figure 12: P_{25} against inscribed diameter. Published including $\sigma_{\eta_{\oplus}}$ 32/53/67/78%, model 32 / 49 / 63 / 73%; excluding, published 6/49/89/99.5%, model 4 / 38 / 79 / 97%.

Hypothesis	Outcome	Evidence
Missing revisits	Real, a cause	Single-visit only; modelling up to six visits raised the uncalibrated yield $1.7\times$.
Characterization-time statistic	Real, a cause	A median over all detectable planets, where the budget needs a mean over those counted; $D^{1.11} \rightarrow D^{1.88}$.
Over-cap characterization charged	Real, a cause	15 000 days charged at 20 pc for an observation never attempted.
Reconstructed target catalog	Real, a cause	HPIC was reachable; the Gaia stand-in lacked 26–34% of distant FGK dwarfs and ran hot.
Coronagraph aperture convention	Real, the cause	Curves are in circumscribed λ/D ; the IWA sat 19% too far out. Yield $14.7 \rightarrow 18.7$.
Collecting area as inscribed circle	Real, partial	Charged the Lyot discard twice.
Characterization at detection phase	Real, a cause	Stark (2019) Sec. 5: the orbit is known, so “the phase of the planet could be optimized.
Characterization budget under albedo draw	Real, a cause	Stark (2024) Sec. 3.2 fixes it at $A_G = 0.2$; albedo penalty $0.050 \rightarrow 0.087$.
Constant T_{sky}	Real, the cause of κ	Stark (2019) Eqs. 5–6 make it a map; with it the uncalibrated yield is 20.92 and κ falls $1.62 \rightarrow 1.000$. Characterization times barely move.
One exozodi draw	Real, fixed	Now 200 draws with the survey re-optimized per draw; exozodi penalty 12.2% (Stark 11.1%).
Split κ (detection vs spectra)	Rejected	A scalar on spectra cannot change the 6/9 m characterization-time ratio.
Exozodi drawn during calibration	Real, a cause	κ fitted to a drawn-exozodi yield cancels the exozodi penalty.
Lognormal exozodi stand-in	Real, a cause	0.2% of stars above 100 zodis vs 21% in HOSTS; penalty $0.7\% \rightarrow 12.2\%$.
Pinned LBTI stars omitted	Real, partial	Worth at most 0.47 EEC here.
$\eta_{\oplus}$ interval read as 86%	Real, the cause of the shape	Fig. 10 requires ln-width ~ 0.8 ; distance $0.142 \rightarrow 0.024$.
Published mode ill-determined	Rejected	Stark’s 498 runs carry 1000 planet draws each; an earlier bootstrap resampled only 498 yields.
Fig. 4 mean compared with the wrong yield	Real, not the cause	Fig. 4 is sampling only at $A_G = 0.2$; the check is now marked tuned.
T_{sky} missing; exozodi law	Real, not the cause	Backgrounds $1.48\times$ too bright; constant-surface-brightness law. Together 1%.
Contrast floor inside the knee	Real, not the cause	Floor applied universally.
Noise-floor screen	Real, not the cause	Admitted nothing above $4.6 L_{\odot}$.
Screening cap, uniform bias, band-pass choice	Rejected	1.3%, destroys 7–9 m, 0.01.
Two detections for orbits	Rejected	Shifts the curve, does not narrow it.
Equal-slope allocation	Rejected	0.00% from an exact optimum.
Blackbody stellar flux	Rejected	+0.019 mag against measured V .
Stark’s per-visit albedo test	Rejected	Gives a weaker penalty (3.3%).
Noise floor, CIC definition, IFS pixels, ordering, dust colour	Rejected	Sec. ??.

Table 4: The ledger. The traps that recur — compensating errors, comparing against the wrong published curve, biased diagnostic sub-samples, and read-off values — are documented in `doc/handoffToNextAgent.md`.

Stark et al. (2024) Fig. 25



This model

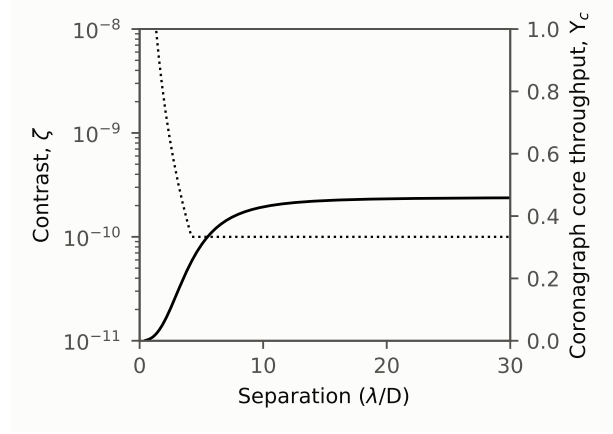


Figure 13: DMVC6 raw contrast (dotted) and core throughput (solid) against separation in circumscribed λ/D . The model curves are the digitized published ones, so this pair checks the digitization, not the model. The model's contrast is shown with the 10^{-10} floor applied, as the exposure times use it; the published curve is the raw simulation. The published figure also shows the PIAA-FPM2.5 (red), which is not used.